
Linear Algebra

What do linear transformations do?

All vectors from -2 to 2

```
In[ ]:= M = Tuples[{-2, -1, 0, 1, 2}, 2]
```

```
Out[ ]:= {{-2, -2}, {-2, -1}, {-2, 0}, {-2, 1}, {-2, 2}, {-1, -2}, {-1, -1},  
          {-1, 0}, {-1, 1}, {-1, 2}, {0, -2}, {0, -1}, {0, 0}, {0, 1}, {0, 2}, {1, -2},  
          {1, -1}, {1, 0}, {1, 1}, {1, 2}, {2, -2}, {2, -1}, {2, 0}, {2, 1}, {2, 2}}
```

```
In[ ]:= A = RandomReal[{-1, 1}, {2, 2}]
```

```
Out[ ]:= {{0.841991, -0.194592}, {0.0835509, -0.650553}}
```

```
In[ ]:= MatrixForm[A]
```

```
Out[ ]//MatrixForm=  

$$\begin{pmatrix} 0.841991 & -0.194592 \\ 0.0835509 & -0.650553 \end{pmatrix}$$

```

```
In[ ]:= Dimensions[M]
```

```
Out[ ]:= {25, 2}
```

M dot matrix

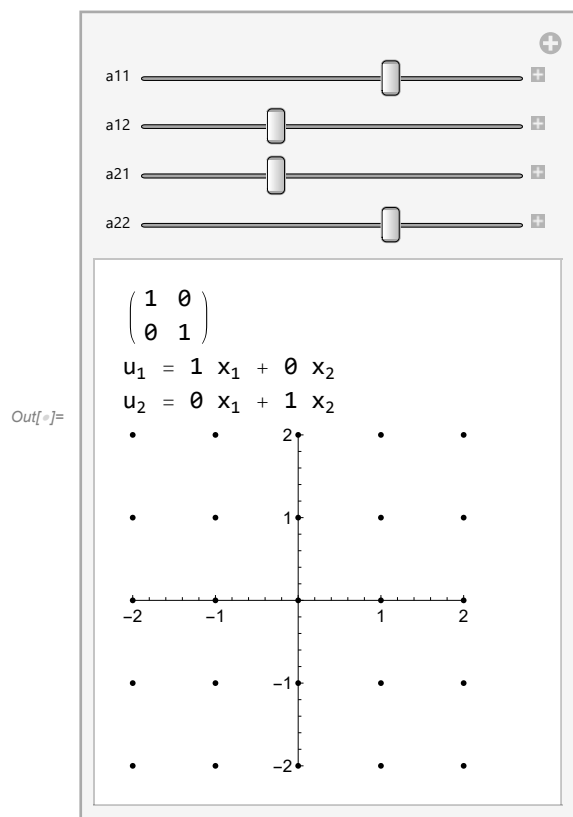
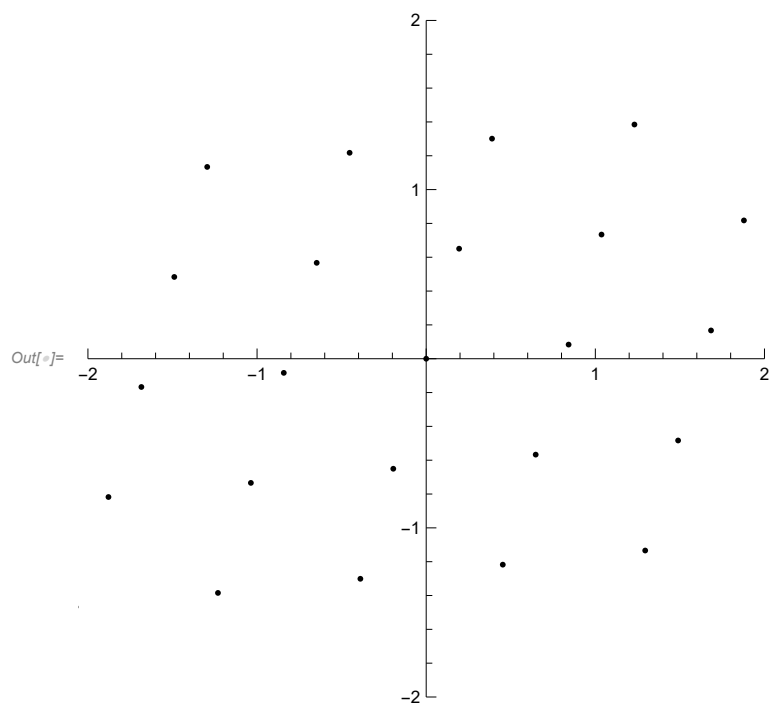
normally matrices that do linear transformations go on the left like

A dot v

matrix dot v

M dot Transpose[A]

In[]:= Graphics[Point[M.Transpose[A]], Axes → True, PlotRange → 2]



$$\text{In}[*]:= \{\{1, 0\}, \{1, 1\}\} \cdot \{-2, 3\}$$

$$\text{Out}[*]= \{-2, 1\}$$

Rigid motions with matrices

Translation

Rotation

Sigma Notation

$$\sum_{i=1}^n a_i$$

$$a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$$

$$\sum_{i=1}^n i$$

$$1 + 2 + 3 + \dots + (n-1) + n$$

$$\sum_{i=1}^6 i = 1 + 2 + 3 + 4 + 5 + 6$$

Vectors

A vector is a list of numbers.

Dot product

Multiply term by term and add them together.

$$\sum_{i=1}^n v_i w_i$$

Cross product

$$\mathbf{v} \times \mathbf{w}$$

makes sense in 3D

It gives a vector perpendicular to v and to w .

Can be expressed as a determinant

Matrices

Out[]//MatrixForm=

$$\begin{pmatrix} 1 & -9 & -5 & 0 & -4 \\ -5 & 8 & -3 & 9 & -4 \\ 2 & 3 & -3 & -7 & 7 \\ -1 & -6 & 8 & -6 & -4 \end{pmatrix}$$

Matrices have two indices.

Row and Column.

Matrix Multiplication

$$\sum_{j=1}^n a_{ij} b_{jk}$$

Square Matrices have the same number of rows and columns.

Out[]//MatrixForm=

$$\begin{pmatrix} 1 & 6 & -2 & 1 & -7 \\ -4 & -8 & -7 & -8 & 2 \\ 9 & -2 & 8 & 4 & -6 \\ -3 & 2 & -9 & 6 & 9 \\ -6 & 2 & -6 & 3 & 6 \end{pmatrix}$$

Tensors

$$a_{ijk}$$

Vector Dynamics

Vectors can change with time.

One type of dynamic is a recursive function that is linear.

$$x_{n+1} = 0.3 y_n$$

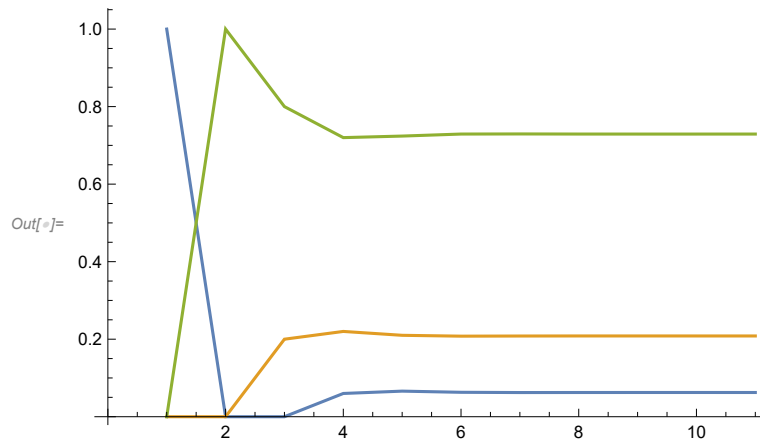
$$y_{n+1} = 0.3 y_n + 0.2 z_n$$

$$z_{n+1} = x_n + 0.4 y_n + 0.8 z_n$$

$$x_1 = 1$$

$$y_1 = 0$$

$$z_1 = 0$$



Random Walks

Adjacency Matrix

$a_{ij} = 1$ if there is an edge from v_j to v_i

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$v_2 \rightarrow v_1$$

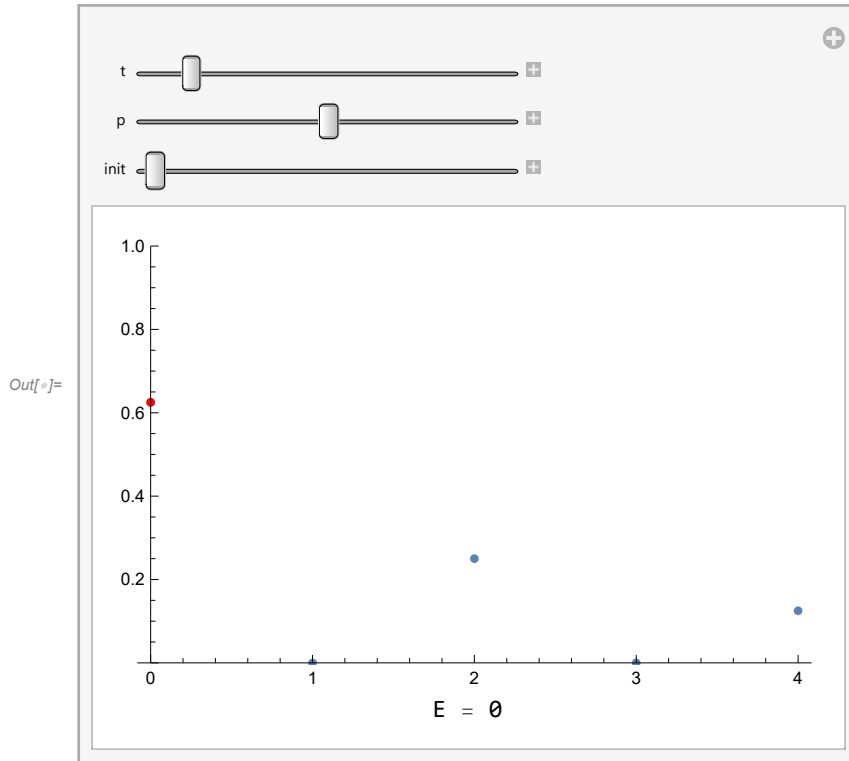


$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ is not a probability vector}$$

$$v_2 \rightarrow v_1 \text{ and } v_2 \rightarrow v_2$$

So we have to divide the column by the total.

Gambler's Ruin



Implementing a Random Walk

```
In[*]:= RWalk[A_][init_, t_] := NestList[RandomChoice[A[[All, #]] → Range[Length[A]]] &, init, t]
```

Visualizing a probability vector on a graph

```
In[*]:= A0 = {{0, 0, 0, 0, 0, 1, 0, 1}, {0, 0, 0, 0, 1, 0, 0, 0},
             {0, 0, 0, 0, 1, 1, 1, 0}, {0, 0, 0, 0, 1, 1, 0, 1}, {1, 1, 0, 1, 0, 0, 0, 0},
             {0, 0, 1, 0, 0, 0, 0, 0}, {1, 1, 0, 1, 0, 0, 0, 0}, {1, 0, 0, 1, 0, 0, 0, 0}};
```

```
In[*]:= A = Transpose[Map[# / Total[#] &, Transpose[A0]]];
```

```
In[*]:= eigen1 = Eigenvectors[A][[2]]
```

```
Out[*]:= {13/10, 2/5, 12/5, 17/10, 6/5, 12/5, 6/5, 1}
```

```
In[*]:= prob = eigen1 / Total[eigen1]
```

```
In[*]:= vertexcolors = Table[i → Blend[{Blue, Red}, 5 N[prob][[i]]], {i, 1, 8}]
```

```
Out[*]:= {1 → Blue, 2 → Blue, 3 → Red, 4 → Red, 5 → Blue, 6 → Red, 7 → Blue, 8 → Blue}
```

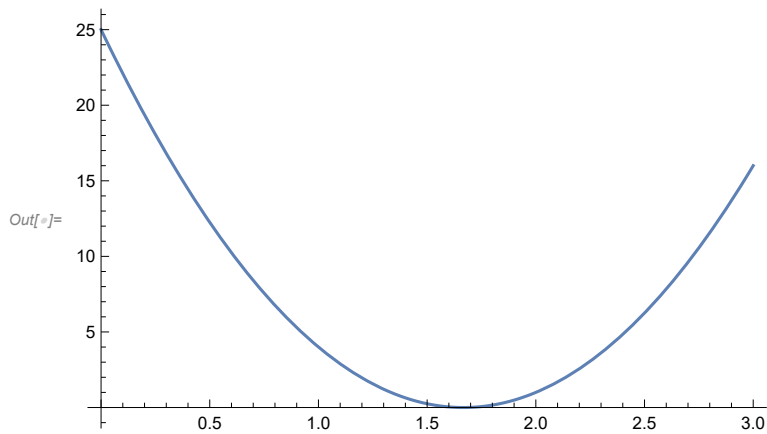
```
In[*]:= AdjacencyGraph[A0, EdgeStyle → Black, VertexStyle → vertexcolors, VertexSize → .5]
```

```
In[*]:= vertexsizes = Table[i → N[prob][[i]], {i, 1, 8}]
```

```
In[ ]:= AdjacencyGraph[A0, VertexSize → vertexsizes]
```

Parabolas

```
In[ ]:= Plot[(3 x - 5)^2, {x, 0, 3}]
```



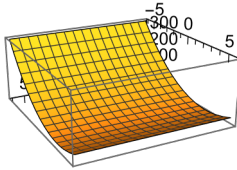
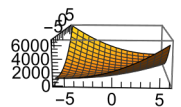
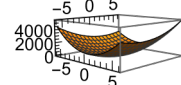
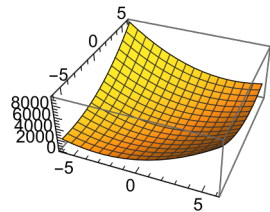
```
In[ ]:= Grid[Table[{MatrixForm[#], MatrixForm[#2], MatrixForm[#{x, y}],
  MatrixForm[#{x, y} - #2], MatrixForm[ExpandAll[#{x, y} - #2]^2]},
  poly[n] = Expand[Total[#{x, y} - #2]^2]] &[
  RandomInteger[{-9, 9}, {n, 2}], RandomInteger[{-9, 9}, n]], {n, 4}], Frame → All]
```

Out[]:=

$\begin{pmatrix} 3 & 8 \end{pmatrix}$	$\begin{pmatrix} -4 \end{pmatrix}$	$\begin{pmatrix} 3x + 8y \end{pmatrix}$	$\begin{pmatrix} 4 + 3x + 8y \end{pmatrix}$	$\begin{pmatrix} 16 + 24x + 9x^2 + 64y + 48xy + 64y^2 \end{pmatrix}$	$16 + 24x + 9x^2 + 64y + 48xy + 64y^2$
$\begin{pmatrix} 9 & -3 \\ -4 & 2 \end{pmatrix}$	$\begin{pmatrix} -1 \\ -1 \end{pmatrix}$	$\begin{pmatrix} 9x - 3y \\ -4x + 2y \end{pmatrix}$	$\begin{pmatrix} 1 + 9x - 3y \\ 1 - 4x + 2y \end{pmatrix}$	$\begin{pmatrix} 1 + 18x + 81x^2 - 6y - 54xy + 9y^2 \\ 1 - 8x + 16x^2 + 4y - 16xy + 4y^2 \end{pmatrix}$	$2 + 10x + 16x^2 - 2y - 10xy + 10y^2$
$\begin{pmatrix} 4 & -3 \\ -3 & 9 \\ 4 & -7 \end{pmatrix}$	$\begin{pmatrix} -2 \\ 5 \\ -3 \end{pmatrix}$	$\begin{pmatrix} 4x - 3y \\ -3x + 9y \\ 4x - 7y \end{pmatrix}$	$\begin{pmatrix} 2 + 4x - 3y \\ -5 - 3x + 9y \\ 3 + 4x - 7y \end{pmatrix}$	$\begin{pmatrix} 4 + 16x + 16x^2 - 12y - 24xy + 9y^2 \\ 25 + 30x + 9x^2 - 90y - 54xy + 81y^2 \\ 9 + 24x + 16x^2 - 42y - 56xy + 49y^2 \end{pmatrix}$	$38 + 70x + 16x^2 - 144y - 144xy + 144y^2$
$\begin{pmatrix} -3 & 8 \\ 5 & 8 \\ -1 & 1 \\ 7 & 4 \end{pmatrix}$	$\begin{pmatrix} 5 \\ -1 \\ 8 \\ -5 \end{pmatrix}$	$\begin{pmatrix} -3x + 8y \\ 5x + 8y \\ -x + y \\ 7x + 4y \end{pmatrix}$	$\begin{pmatrix} -5 - 3x + 8y \\ 1 + 5x + 8y \\ -8 - x + y \\ 5 + 7x + 4y \end{pmatrix}$	$\begin{pmatrix} 25 + 30x + 9x^2 - 80y - 48xy + 64y^2 \\ 1 + 10x + 25x^2 + 16y + 80xy + 64y^2 \\ 64 + 16x + x^2 - 16y - 2xy + y^2 \\ 25 + 70x + 49x^2 + 40y + 56xy + 16y^2 \end{pmatrix}$	$115 + 12x + 12x^2 + 40y + 80xy + 16y^2$

```
In[ ]:= Grid[
  Table[{N[Minimize[poly[n], {x, y}]], Plot3D[poly[n], {x, -6, 6}, {y, -6, 6}]], {n, 4}},
  Frame → All]
```

Out[]:=

$\{0., \{x \rightarrow 0., y \rightarrow 3.\}\}$	
$\{0., \{x \rightarrow -0.162162, y \rightarrow -0.351351\}\}$	
$\{0.971332, \{x \rightarrow -0.623103, y \rightarrow 0.306071\}\}$	
$\{58.6649, \{x \rightarrow -0.314136, y \rightarrow -0.931937\}\}$	

```
In[ ]:= Grid[Table[{MatrixForm[#], MatrixForm[#2], MatrixForm[#{x, y, z}],
  MatrixForm[#{x, y, z} - #2], MatrixForm[ExpandAll[(#{x, y, z} - #2)^2]],
  poly3d[n] = Expand[Total[(#{x, y, z} - #2)^2]]} &[
  RandomInteger[{-9, 9}, {n, 3}], RandomInteger[{-9, 9}, n]], {n, {4}}], Frame → All]
```

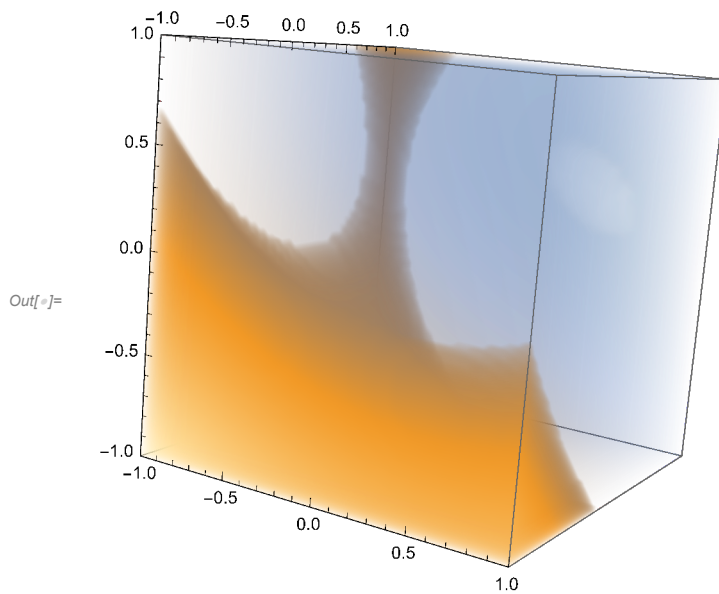
Out[]:=

$\begin{pmatrix} -6 & 7 & 2 \\ 6 & -9 & -3 \\ -7 & 3 & -8 \\ 2 & -5 & 1 \end{pmatrix}$	$\begin{pmatrix} -8 \\ -8 \\ -3 \\ 2 \end{pmatrix}$	$\begin{pmatrix} -6x + 7y + 2z \\ 6x - 9y - 3z \\ -7x + 3y - 8z \\ 2x - 5y + z \end{pmatrix}$	$\begin{pmatrix} 8 - 6x + 7y + 2z \\ 8 + 6x - 9y - 3z \\ 3 - 7x + 3y - 8z \\ -2 + 2x - 5y + z \end{pmatrix}$	$\begin{pmatrix} 64 - 96x + 36x^2 + 112y - 84xy + 49y^2 \\ 64 + 96x + 36x^2 - 144y - 108xy + 81y^2 \\ 9 - 42x + 49x^2 + 18y - 42xy + 9y^2 \\ 4 - 8x + 4x^2 + 20y - 20xy + 25y^2 \end{pmatrix}$
--	---	---	--	--

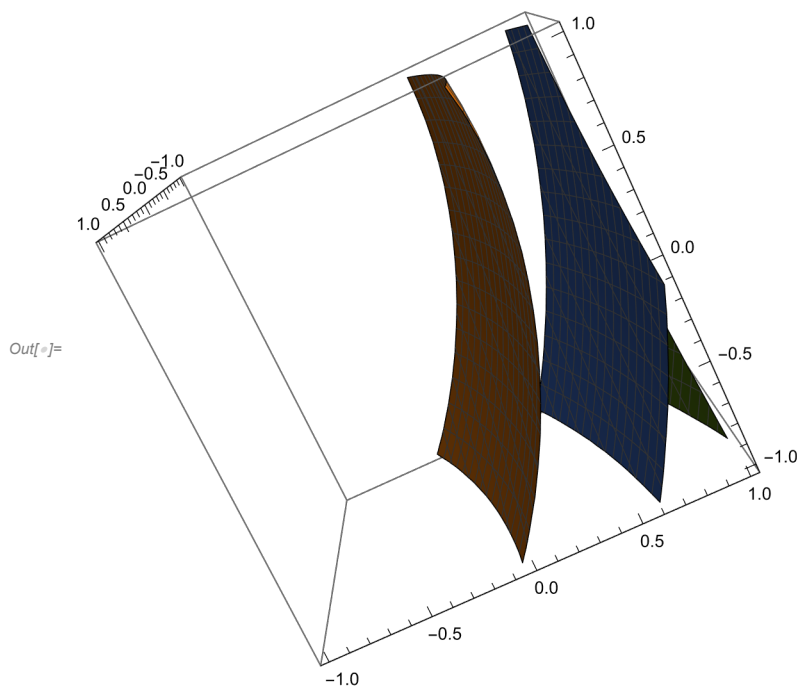
```
In[ ]:= N[Minimize[poly3d[4], {x, y, z}]]
```

Out[]:= $\{1.83837, \{x \rightarrow 0.663084, y \rightarrow 0.243828, z \rightarrow 0.448819\}\}$

```
In[ ]:= DensityPlot3D[poly3d[4], {x, -1, 1}, {y, -1, 1}, {z, -1, 1}]
```



```
In[ ]:= ContourPlot3D[poly3d[4], {x, -1, 1}, {y, -1, 1}, {z, -1, 1}]
```



Text generation

Our text generator is based on the text found at
<http://www.gutenberg.org/ebooks/35>

The Time machine

by H.G. Wells

The next letter depends on the letter before it

In the text, there are 2944 “in” and 1344 “it” and 1151 “i “....

	i																				
Out[]:=	2944	1344	1151	803	494	486	443	380	372	350	325	227	171	160	111	91	90	88	46	16	16
	n	t		s	l	m	o	c	g	r	d	e	v	f	a		k	b	p	'	u

These values form a vector.

```
Out[ ]:= {2944, 1344, 1151, 803, 494, 486, 443, 380, 372, 350, 325,
          227, 171, 160, 111, 91, 90, 88, 46, 16, 16, 15, 14, 12, 9, 6, 3, 1, 1}
```

```
In[ ]:= vector = {2944, 1344, 1151, 803, 494, 486, 443, 380, 372, 350,
                  325, 227, 171, 160, 111, 91, 90, 88, 46, 16, 16, 15, 14, 12, 9, 6, 3, 1, 1};
```

```
In[ ]:= Total[vector]
```

```
Out[ ]:= 10169
```

To make a probability vector it has to add up to one.

```
In[ ]:= probabilityvector = N[vector / 10169]
```

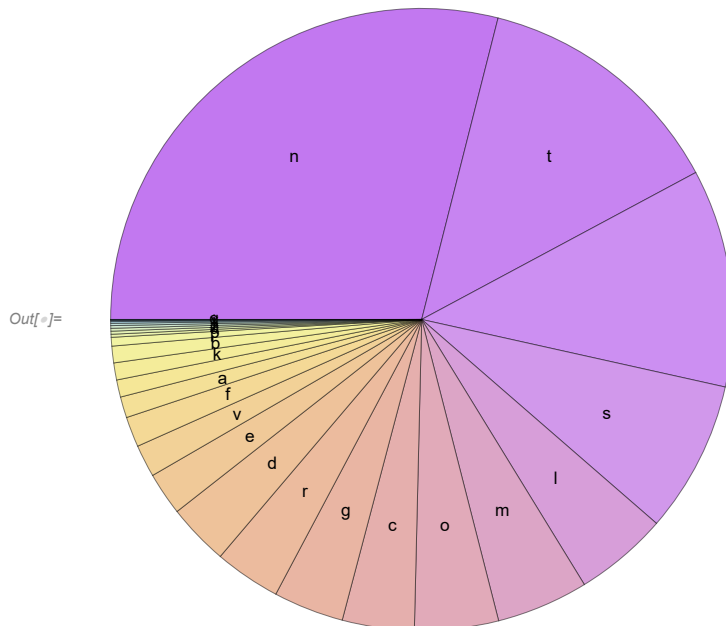
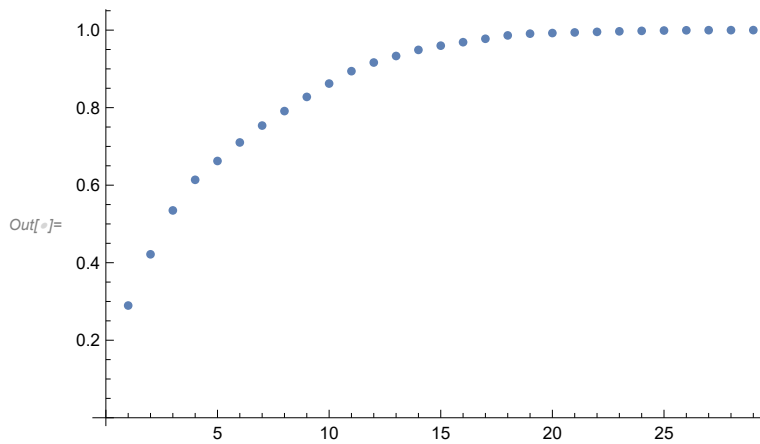
```
Out[ ]:= {0.289507, 0.132166, 0.113187, 0.0789655, 0.048579, 0.0477923, 0.0435638, 0.0373685,
          0.0365818, 0.0344183, 0.0319599, 0.0223227, 0.0168158, 0.0157341, 0.0109155, 0.00894877,
          0.00885043, 0.00865375, 0.00452355, 0.00157341, 0.00157341, 0.00147507, 0.00137673,
          0.00118006, 0.000885043, 0.000590029, 0.000295014, 0.0000983381, 0.0000983381}
```

Find the thresholds by accumulating the values. When we pick a random number between 0 and 1, the thresholds show us which one to pick.

```
In[ ]:= Accumulate[probabilityvector]
```

```
Out[ ]:= {0.289507, 0.421674, 0.534861, 0.613826, 0.662405, 0.710198,
          0.753761, 0.79113, 0.827712, 0.86213, 0.89409, 0.916413, 0.933228, 0.948963,
          0.959878, 0.968827, 0.977677, 0.986331, 0.990855, 0.992428, 0.994001,
          0.995476, 0.996853, 0.998033, 0.998918, 0.999508, 0.999803, 0.999902, 1.}
```

In[]:= **ListPlot[Accumulate[probabilityvector]]**



In[]:= **Manipulate[BlockRandom[SeedRandom[seed];**
Column[{#, TopChoices[#]}] &[Nest[MarkovTimeText, "", t]]],
{seed, 1, 100, 1}, {t, 0, 100, 1}, SaveDefinitions -> True]

Using the last two letters

In[]:= **Manipulate[BlockRandom[SeedRandom[seed];**
Column[{#, Style[FullForm[StringTake[#, -Min[2, StringLength[#]]]], Bold, Red],
TopChoices3[#]}] &[Nest[MarkovTimeText3, "", t]]],
{seed, 1, 100, 1}, {t, 0, 100, 1}, SaveDefinitions -> True]

In[]:= **Total[{373, 122, 72, 71, 28, 27}]**

Out[]:= 693

```
In[ ]:= {373, 122, 72, 71, 28, 27} / 693
```

```
Out[ ]:=  $\left\{ \frac{373}{693}, \frac{122}{693}, \frac{8}{77}, \frac{71}{693}, \frac{4}{99}, \frac{3}{77} \right\}$ 
```

```
In[ ]:= N[{373, 122, 72, 71, 28, 27} / 693]
```

```
Out[ ]:= {0.53824, 0.176046, 0.103896, 0.102453, 0.040404, 0.038961}
```

```
In[ ]:= Accumulate[N[{373, 122, 72, 71, 28, 27} / 693]]
```

```
Out[ ]:= {0.53824, 0.714286, 0.818182, 0.920635, 0.961039, 1.}
```

```
In[ ]:= 0.5382395382395382` + 0.17604617604617603`
```

```
Out[ ]:= 0.714286
```

```
In[ ]:= 0.7142857142857142` + 0.1038961038961039`
```

```
Out[ ]:= 0.818182
```

{, t, h, e, c, i}

Group Activity

```
In[ ]:= Manipulate[Column[{s, Style[FullForm[StringTake[s, -Min[2, StringLength[s]]]], Bold, Red],
  If[show, TopChoices3[s], Column[Table["XXX", 3]]], RandomReal[]}],
  {{s, "to be"}}, {show, {False, True}}, SaveDefinitions -> True]
```

Using last 7 letters

```
In[ ]:= Manipulate[BlockRandom[SeedRandom[seed];
  Nest[MarkovTimeText8, StringTake[text, 18], n]],
  {seed, 1, 100, 1}, {n, 0, 1500, 1}, SaveDefinitions -> True]
```